# QQ Super Compression

User Manual / 1.4.20 · English

## Control dynamics. Keep the attack natural.

For vocals, instruments, buses and mastering. The dynamics core has no conventional Attack or Release knobs. Shape the sound with Lookahead, Transparent, Ratio, Threshold and Mix; the Limiter has three separate Ceiling recovery modes.

Classic controls dynamics with a fixed ratio in dB; Super uses the original QQ curve. Compressor offers single, upward and dual compression. Clipper provides downward compression only, while Limiter adds a Ceiling for final peak control. In ST, L/R Link and M/S Link adjust left/right or mid/side coupling with one set of parameters. The Quick Start introduces both controls. Adjust moves the graph without changing the sound.




**Start with what you hear**
Decide how steady the dynamics should be and how much articulation to retain. The Display helps explain changes; it does not replace listening.

# Quick Start and reading guide

Start with Compressor / ST, then choose another mode if needed.

1. **Choose a starting point**
   CLASSIC, SINGLE, ST, SC: INT; Lookahead 25.3 ms, Window 25.3 ms, Transparent off and OS 1x.

2. **Meet the two ST Links**
   L/R Link and M/S Link both default to 100%. Lower the first to blend in independent left/right compression, or the second for independent mid/side compression. You still use one set of knobs. See pages 21-22.

3. **Set the strength and amount**
   Use Threshold to choose where processing begins. Increase Ratio from 1:1, then use Mix to set the amount.

4. **Compare at matched loudness**
   Play representative material, click Match, then listen to attacks, tails and placement in the mix.

5. **For stronger downward compression**
   Read the Clipper and Limiter workflows. Understand L/R Link first, then compare Window, loudness and dynamics.

6. **Check the complete mix**
   Use Bypass or A/B to compare. A louder result is not necessarily a better one.


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# Know the interface

The same audio controls remain in place across all themes.




Light / ST. The values illustrate the interface; they are not a recommended preset.

| Area | Use it to |
|---|---|
| Display | View historical level and gain references under current settings; see page 16. |
| Threshold / Range faders | Threshold starts compression above its level; Range stops it at or above its level. These two faders select the processing region in normal Single mode; see page 8. |
| Input / Output / Gain +/- | Check input, output and effective dynamic gain. |
| Bottom knobs and SINGLE / DUAL | Adjust moves the graph; the other knobs control level, Ratio and Mix. Switch Single / Dual beside Ratio. |
| Four rows at the lower right | Mode and the two ST Link values, followed by Transparent, Lookahead + Window, then Monitor + OS. Clipper / Limiter use L/R Link in the first row and have no channel Monitor. |
| Top buttons | Cycle through COMPRESSOR / CLIPPER / LIMITER, then choose FULL / ECO, sidechain, theme, A/B and Bypass. |

The interface uses a 3:2 landscape aspect ratio. Drag the bottom-right corner to resize it. The plugin remembers the last size when reopened.

# Processing mode, then channel mode

The top button selects Compressor / Clipper / Limiter.




Click the leftmost processing-mode button to cycle Compressor → Clipper → Limiter.

The processing mode determines which features are available. Compressor handles general dynamics. Clipper focuses on downward compression at higher ratios. Limiter adds a Ceiling after dynamics processing to control final peaks.

| Processing mode | Main uses and controls |
|---|---|
| Compressor | Upward, downward or dual compression. Offers ST / LR / MS; ST has two numeric Link controls for coupling. |
| Clipper | Downward only, Ratio 10:1-1000:1. One set of parameters and L/R Link, without dual compression or Ceiling. |
| Limiter | One set of parameters and L/R Link. Offers single / dual compression, Ceiling, TP and level-matched monitoring. |

## Each mode remembers its parameters

Switch from Compressor to Clipper and back, and your previous Compressor settings return. Each A/B snapshot also stores all three sets of mode parameters. See page 44.

## Mode is a separate choice

In Compressor, the lower-right Mode button cycles ST → LR → MS → ST. ST uses one set of parameters; LR lets you adjust left and right separately; MS lets you adjust mid and side separately. This is a different choice from the top processing-mode button.

> **Adjust coupling directly in ST**
> ST L/R Link and M/S Link both default to 100%. Lower either value to try more independent processing without switching to the two sets of knobs in LR / MS. See pages 21-22.

# Themes and everyday controls

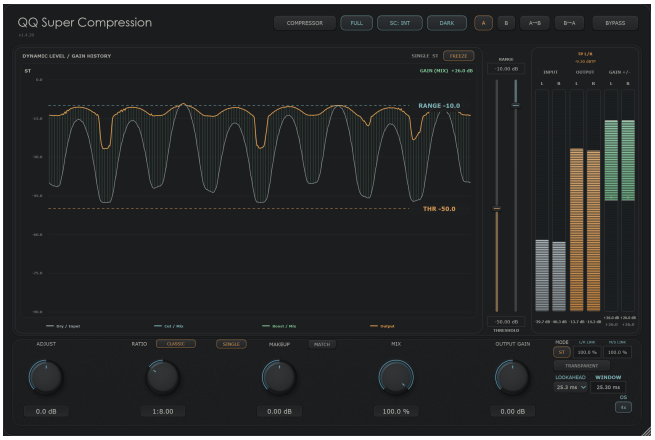
Click the top-right button labeled LIGHT, DARK or CLASSIC to cycle themes. The last selection is restored when the plugin opens again.




The theme button is between SC and A/B.

| Theme | Appearance |
|---|---|
| Light | Light interface. |
| Dark | Subdued dark interface. |
| Classic | Original classic dark interface. |

All three themes change appearance only. The CLASSIC theme and the CLASSIC algorithm in the Ratio section are independent: one changes the colors, the other changes the processing curve.




Dark




Classic

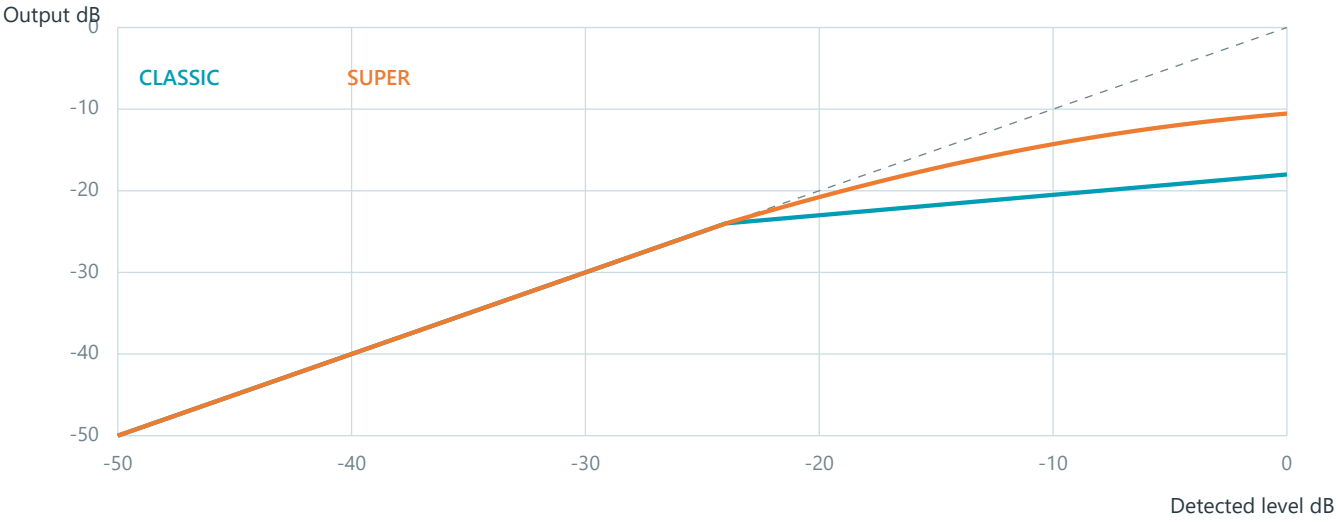
| Adjustment | Result |
|---|---|
| Drag / Shift-drag | Normal / fine adjustment; press or release Shift while dragging. |
| Double-click a value | Type an exact value. |
| Alt / Option + left click | Restore the control's reset value; see page 40 for thresholds. Linked controls follow within their shared limits. |
| Ctrl/Cmd + Z | Undo; add Shift to redo. |
| Display wheel | Wheel: vertical zoom; Shift: Adjust; Ctrl: horizontal zoom. Selection and FREEZE: page 16. |

Hover over a control for Chinese help. Parameter labels retain their displayed names. On macOS, use Option wherever this manual says Alt.

# Classic and Super: two dynamic curves

Select the algorithm in the lower Ratio area. DUAL has separate UP / DOWN choices.

| Algorithm | How it works |
|---|---|
| Classic | A fixed dB ratio above the threshold. For downward compression, divide the excess dB by Ratio to find how much excess remains. |
| Super | The original QQ curve. Ratio sets its strength, but does not describe a fixed dB compression ratio at every detector level. |




Static curves: Threshold -24 dB, Ratio 4:1, processing above threshold; no gain compensation, level transitions or detector timing.

## A worked example

Classic downward compression: Threshold = -10 dB, Ratio = 5:1. In this example, audio above Threshold remains eligible for compression. A steady detector level of 0 dB is 10 dB above Threshold; compression retains 2 dB of that excess, giving **-8 dB**. A level of -12 dB is below Threshold and is not attenuated.

This example uses internal detection, 0 dB Makeup and Output Gain, and Mix=100%. The formula describes a steady detector level, not a hard limit on individual samples. For changing signals and the Display, see pages 16-17.

## Upward compression: lift audio above Threshold

Classic and Super both support upward compression. Set Ratio below 1:1 to lift audio above Threshold. Their gain curves differ, but both use Threshold and Range to select the audio to process; see pages 8-9.

> **Threshold range**
> Classic and Super support -inf (negative infinity) to +24 dB for Threshold, UP and DOWN. Existing -90 / 0 dB defaults remain; see page 40 for -inf. Adjust and zoom change the view, not threshold values.

New Compressor instances remember the last manually selected algorithm; projects and A/B restore their saved choice. Clipper and Limiter keep separate settings. A short crossfade reduces transients when changing algorithms.

# Ratio and Mix: strength, then amount

Adjust changes only the graph position; see page 13.




## Ratio

Normal Single processing is neutral at 1:1: above it, processing is downward; below it, processing is upward. Within the active threshold range, higher downward ratios usually cut more, while smaller upward ratios usually boost more. Dual separates UP RATIO and DOWN RATIO; see page 10.

See page 6 for the meaning of Ratio in Classic and Super. Normal Single mode ranges from 1:200 to 200:1, with 1:1 as the default. Final peak protection comes from the Limiter Ceiling; increasing the normal compression Ratio alone does not enable Limiter.

## Mix

Mix blends Dry with processed Wet. It is also a useful processing amount control: first make the boost or cut clear, then lower Mix to blend back the original dynamics you want to retain.

> **Dynamic gain includes Mix**
> At Mix = 0%, effective dynamic gain is 0 dB; at 100%, the full processing amount is retained. Intermediate values are not simply the gain in dB multiplied by Mix. Makeup and Output Gain are excluded from this dynamic gain.

A practical order: find the compression character with Ratio, set its final strength with Mix, then compare loudness. Use Adjust if the graph is too high or too low; it does not affect the sound.

# Single: Threshold and Range

Choose which levels to process, then set the amount.

## Threshold: compression starts above this level

Threshold sets the level where processing begins. At or below it, there is no compression; above it, Ratio determines downward attenuation or upward gain. Its range is -inf to +24 dB; Comp Single defaults to -90 dB.

## Range: compression stops at this level

Range preserves the dynamics of louder audio. At or above Range, neither attenuation nor upward gain is applied. Compression therefore acts only between Threshold and Range. In normal Single mode, the Range fader is above Threshold. Its finite range is -90 to +24 dB.

## How to use it

For example, set Threshold to -30 dB and Range to -10 dB:

| Signal level | What happens |
|---|---|
| -30 dB or below | No compression. |
| Above -30 dB, below -10 dB | Attenuate or lift according to Ratio. |
| -10 dB or above | No compression; preserve these dynamics. |

Lower Range to exclude more loud audio from compression; raise it to include louder audio. Keep Threshold fixed first, then move Range and listen for the parts that regain their original dynamics.

## Range OFF: disable the Range control

Move Range to OFF at the top to disable its range restriction. Downward processing still follows Threshold and Ratio; see page 9 for the upper end of upward gain. OFF disables Range, while compression remains active.

> **Special cases**
> Range = 0 dB stops compression at 0 dB. Range OFF removes this stopping level.
> When Range and Threshold coincide, no processing interval remains and dynamic processing stops. The faders move together when they meet; drag back to separate them.
> Processing tapers off near Range and stops at it. This transition follows signal level; it adds no Attack / Release stage.

**If you hear crackles:** Range can produce crackles on some material. Try enabling Transparent for more stable detection, then compare. Transparent may also attenuate audio before and after loud events, so listen for both effects; see pages 19-20.

On this page, 'no compression' means no dynamic cut or boost. Makeup, Mix and Output Gain still apply. EXT uses the external sidechain level for these decisions. Clipper and Limiter do not use Range; see pages 30-33.

# Downward and upward compression

First understand the gain change, then compare after matching levels.

## Downward compression: attenuate audio above Threshold

In SINGLE, turn Ratio right from 1:1 to values such as 2:1, 4:1 or 8:1. Audio above Threshold is attenuated, reducing the difference between loud and quiet levels. Audio at or below Threshold is not attenuated.

The main sound may become quieter. Raising Makeup or Output to match its original level also raises the uncompressed audio below Threshold.

## Upward compression: lift audio above Threshold

Turn Ratio left from 1:1 to values such as 1:2, 1:4 or 1:8 for upward compression. Within the active region, quieter audio generally receives more lift than louder audio, reducing the dynamic difference. Audio at or below Threshold is not lifted.

Classic and Super both support these directions. Single upward compression uses finite Range as the upper end of its lift. With Range OFF, lift ends at 0 dB; audio at or above that level is not boosted. Dual uses DOWN as the upper end of upward gain.

## Compare before adding level compensation

| Level region | Downward compression | Upward compression |
|---|---|---|
| At or below Threshold | No attenuation | No lift |
| Active region above Threshold | Attenuate to reduce dynamic differences | Lift to reduce dynamic differences |
| At or above an active Range | No attenuation | No lift |

## The symmetry between upward and downward compression

Within the same algorithm, level-matching audio above Threshold can make downward 8:1 and upward 1:8 produce the same result in the main processing region. Classic and Super each have this symmetry. Overall compensation also affects audio below Threshold, where the results differ; see page 15 for numbers.

> **Choosing between them**
> To raise quiet audio below Threshold as well, try downward compression followed by compensation. To lift selected details while retaining quieter noise, breaths or ambience, try upward compression with Threshold above the sounds you want to leave alone. Compare in the full mix.

Upward gain fades in near Threshold; processing tapers off near Range. Do not expect exact equivalence around those boundaries. Changing Mix also calls for a fresh listening comparison.

# Dual: lift quieter parts, control louder ones

Click the small button beside Ratio to change SINGLE to DUAL.




DUAL: UP RATIO and DOWN RATIO control the two directions separately. UP / DOWN thresholds replace Threshold / Range.

| **Original level** | **Boosted by UP RATIO** | **Cut by DOWN RATIO** |
|---|---|---|
| At or below UP | Above UP, below DOWN | Above DOWN |

UP threshold — DOWN threshold

Detected level: low to high (processing regions)

At DOWN, dynamic gain returns to 0 dB. Above it, upward processing stops and only Down Ratio applies. The two gains never stack at the same detector level. Dual has no Range control.

| Starting a new instance | How to read it |
|---|---|
| UP: -90 dB. DOWN: 0 dB. The defaults are the same for Classic and Super. | The thresholds start at opposite ends; this does not disable upward processing. |
| Up Ratio = 1:1; Down Ratio = 1:1 | Neither direction changes dynamics until Ratio is adjusted. |

**Threshold order and collision**
DOWN must stay at or above UP. When one fader reaches the other, it pushes the other along. Dynamic processing stops when the thresholds coincide.

# Dual switches: audition each direction

UP / DOWN each have an on/off switch and an algorithm button for separate auditioning and curve selection.




Upward and downward compression are on. Each algorithm is shown in its Ratio area. The switches preserve Ratio values.

| UP / DOWN | Processing you hear |
|---|---|
| ON / ON | Boost between UP and DOWN; attenuation above DOWN. |
| ON / OFF | Upward only. Boost still stops at DOWN. |
| OFF / ON | Downward only. No dynamic processing below DOWN. |
| OFF / OFF | No dynamic boost or attenuation. Other gain and Mix settings still apply. |

1. **Set up the full Dual sound**
   Set both thresholds and Ratios. Play material with quiet and loud passages.

2. **Hear what each direction contributes**
   Turn DOWN off to hear upward compression, then UP off to hear downward compression, then enable both. Each direction can use Classic or Super; changing the UP algorithm does not change DOWN.

3. **Check the display and compare at similar loudness**
   The switches update dynamic gain and the Display history reference. Compensate overall level, then compare detail, articulation and consistency.

> **Each switch affects only its own compression direction**
> Each ON / OFF switch uses a short crossfade to reduce transients. Disabling one direction does not move thresholds or extend the other into its region. Normal LR / MS allows separate channel switches. With Mode LINK enabled, corresponding UP switches follow each other, as do DOWN switches; UP and DOWN remain independent.

# Ratio LINK: opposite proportional changes

Normal DUAL has a small LINK between its two Ratio controls. Limiter does not.




Up and Down retain their existing relative relationship. Enabling LINK does not make values jump.

Ratio LINK defaults to on at first use, then remembers your last choice. When reopening a saved project, the LINK state saved in that project takes priority.

| Initial Down / Up | Adjust Down | Linked Up |
|---|---|---|
| 1:1 / 1:1 | 1:1 → 2:1 | 1:2 |
| 4:1 / 1:2 | 4:1 → 8:1 | 1:4 |
| 8:1 / 1:4 | 8:1 → 4:1 | 1:2 |

## Preserve the relationship, without re-pairing

For example, doubling Down halves the Up value; adjusting Up uses the same inverse relationship. Starting at 4:1 and 1:2 will not force them into a reciprocal pair. Dragging, fine adjustment and direct entry all follow this rule.

> **Linked controls stop together at their limits**
> Normal compression offers Up from 1:200 to 1:1 and Down from 1:1 to 200:1. If either would exceed its range, both stop at the shared limit. Reverse the adjustment to continue. Toggling LINK alone does not change the ratios.

## To hear upward processing alone

Switch DOWN off to preserve its Ratio while auditioning Up. To adjust Up Ratio independently as well, turn off the small Ratio LINK; otherwise the Down value still follows your adjustment.

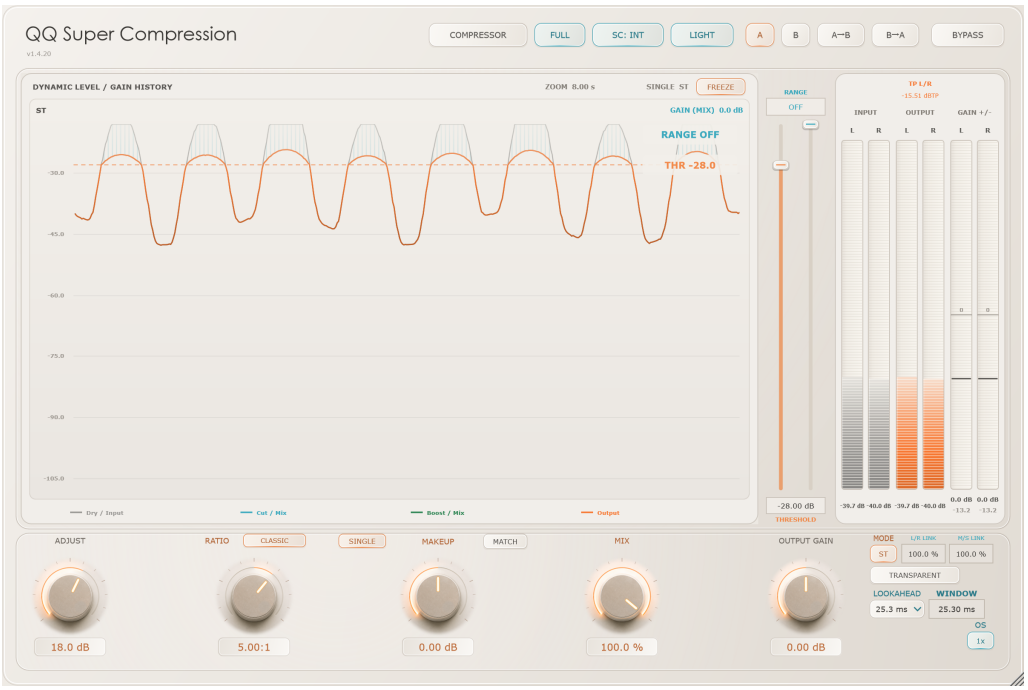
## Also available in LR / MS

In normal LR / MS, the small LINK connects Up and Down ratios within each channel. Mode LINK connects channels; see page 25. Limiter UP / DOWN ratios are always independent: set downward compression first, then add a little upward compression; see page 34.

# Adjust: move only the graph

Reposition the view when waveforms sit too high or too low.

Adjust moves the Display content vertically: waveforms, level scale and threshold lines move together. It does not change input level, Threshold, Ratio or the actual compression result.




Example: Adjust +18 dB moves the graph content upward. Threshold remains -28 dB.

| Action | What you see |
|---|---|
| Increase Adjust | Waveforms, scale and threshold lines move up together. |
| Decrease Adjust | The content moves down; displayed dB values retain their meaning. |
| Shift + wheel on Display | Edits the same Adjust value and updates the knob, while retaining zoom. |

## Range, reset and saving

Adjust ranges from -90 to +90 dB and defaults to 0. Drag vertically or double-click to type. Alt-click (Option on Mac) resets it to 0. Right-clicking Display restores the full view and resets Adjust.

Projects save Adjust. It stays unchanged across A/B and Compressor / Clipper / Limiter changes, and affects neither sound, loudness measurement nor plugin latency.

> **Opening older projects**
> Input Gain and its Input / Output LINK have been removed. Saved Input gain no longer applies; existing Makeup, Output and thresholds keep their values. Recheck projects that used non-zero Input gain. Clipper / Limiter Output LINK remains available; see page 44.

# Match and Makeup: measure from playback start

Click during playback; each click uses the entire current playback up to that point.

| Control | Use |
| --- | --- |
| Makeup | Adjusts Wet level before Mix: Comp ±30 dB; Clipper / Limiter ±120 dB. |
| Output Gain | Comp ±24 dB; Clipper / Limiter ±120 dB. Headphone monitoring: pages 37-38. |
| Match | Each click sets Makeup from cumulative loudness; this is not continuous automatic gain. |

1. **Start playback and establish a measurement**
   Choose the processing mode. Play representative, non-silent material. Once valid loudness data is available, you can click Match during playback.

2. **Click again whenever needed**
   The first click does not clear the measurement. Clicking at 10 and 20 seconds uses the first 10 and first 20 seconds respectively. It does not add the previous compensation again.

3. **A new playback starts a new measurement**
   Stopping retains the result; the next playback starts a new measurement. If playback does not stop, loops and jumps remain part of the same Match measurement.

> **FULL and ECO**
> FULL continues measuring with the editor closed. In ECO, open the editor before playback and keep it open for a complete Match measurement. Missing analysis makes Match unavailable for that playback; restart playback to restore it.

## Fix the comparison conditions before judging the sound

All three modes compare the original input loudness with the dynamically processed Wet signal before Makeup, then write the correction to Makeup. Adjust, Output Gain, Ceiling and headphone monitoring are excluded from Match measurement.

Match does not automatically equalize final output loudness. Mix, Ceiling and monitoring still affect what you hear; trim Makeup if needed. Clicking Match retains the measurement start. To compare only new settings, stop and replay the same passage.

> **Loudness and peaks are separate**
> Match compensates loudness; Ceiling controls final peaks. Check output and TP afterwards, and use Bypass / A/B to compare attacks, tails and dynamics.

# A/B: compare algorithms and directions




A and B hold two sound settings. A→B and B→A copy between them.

1. **Save the first idea**
   Set the algorithm and processing amount in A. Copy A→B, then switch to B. Copying replaces the destination sound settings.

2. **Change the part you want to compare**
   For example, use Classic in A and Super in B, or compare downward and upward processing. Keep Lookahead, detector source and monitoring the same where possible.

3. **Match each, then switch**
   Play the same material for each setting and use Match; trim Makeup if needed. Click A / B and compare the sound, without mistaking louder for better.

| Recalled by A/B | Outside the A/B sound snapshot |
|---|---|
| Algorithms, single / dual mode, thresholds, Ratio, UP / DOWN switches, Makeup, Output, Mix, Lookahead, Transparent, Window, global OS and sidechain; also the active processing mode, TP, Ceiling on/off and level, recovery, Link states and all three sets of processing parameters. | Theme, Adjust, FULL / ECO, FREEZE, 1:1 headphone monitoring, channel solo, Bypass and SC LISTEN. |

## Upward/downward symmetry: 8:1 and 1:8

Compare within Classic or within Super. Use the same Threshold, Lookahead, Window and Transparent state, Range OFF, Mix 100%, and no Ceiling or other extra processing. Match levels using audio above Threshold: the main processing region can then give the same result.

Example: Classic, Threshold = -24 dB. A uses downward 8:1 followed by +21 dB compensation. B uses upward 1:8 with no compensation. Steady-level results:

| Input level | A: Down + compensation | B: Up |
|---|---|---|
| -12 dB (above Threshold) | -1.5 dB | -1.5 dB |
| -6 dB (above Threshold) | -0.75 dB | -0.75 dB |
| -36 dB (below Threshold) | -15 dB | -36 dB |

Neither setting compresses below Threshold, but A's overall compensation raises that audio too, producing a difference. Super has the same type of symmetry with a different compensation amount. This does not mean that Super and Classic sound identical.

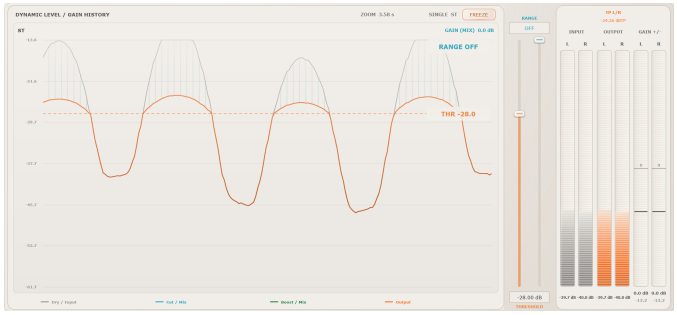
The upward transition near Threshold, an active Range, a different Mix or extreme boosts can break this correspondence. Match measures loudness across the whole playback, including audio below Threshold. To examine this example, align the level of the main sound above Threshold.

> **Switching and latency**
> A/B uses transitions to reduce switching artifacts. Keep Lookahead and OS equal to focus on the sound; changing modes or rates may make the host update delay compensation. A/B recalls each saved Window value and realigns the Display accordingly.

# Display: zoom, pan and Freeze

All three processing modes share these controls. Zoom and pan do not change the sound.

Display shows level and gain references over time. A rectangle zooms both axes; the wheel makes small adjustments to one axis. You can alternate between them.




During playback, the graph keeps the selected scale and follows the newest audio at the right edge.

| Action | Result |
|---|---|
| Left-drag a rectangle | Zoom the selected time and level range. When stopped, inspect the selected passage. |
| Mouse wheel | Scroll up to zoom in or down to zoom out vertically. Inside the graph, centre on the pointer level; above it, keep 0 dB at the top. |
| Shift + wheel | Edit Adjust and its knob together. Scroll up to move content up, or down to move it down, without changing zoom. |
| Ctrl + wheel | Zoom time horizontally. Centre on the pointer when stopped; follow the newest audio during playback. |
| Right-click Display | Restore 8 seconds of history, 0 to -90 dB, and reset Adjust to 0. |

## Freeze: retain the graph after Stop

With FREEZE lit at the upper right, the graph scrolls during playback and freezes after Stop. When off, it keeps updating. The initial default is on; new instances remember your last manual choice. Existing instances remain independent. Projects save their own state; A/B does not change it.

While frozen, you can still zoom, change sound parameters and inspect the updated curves. The meters keep working. Playback resumes live following at the selected scale. Recording, offline export and hosts without transport information keep updating. Reopening the editor in ECO clears the old graph, even with Freeze on.

## Zoom the view, keep the thresholds

The fader scales follow the graph; Threshold, UP, DOWN and Range values remain unchanged. Thresholds outside the view retain their readouts; use Adjust to bring them into view. The -inf marker stays at the bottom edge. Zoom does not change compression or latency.

Grey is input; blue / green indicate cut / boost; orange is output. Curves follow current settings, including Window and HPF drags. HPF UPDATING means calculation is still running. Display is a level reference, not a recording of the final waveform.

Limiter input and output share the same time position; the output reference includes Ceiling when enabled. The external sidechain curve is independent of Window and Transparent; HPF / Key Gain change it.

# Lookahead and Window: latency and tone

Keep latency fixed while comparing the sound of shorter Lookahead lengths.

Lookahead reserves time to read ahead. The plug-in delays output so it can analyze upcoming audio; this sets the base latency and the maximum Window. Window sets how far before and after the current sample the processor looks for peaks when calculating gain.

| Control | What it sets |
| --- | --- |
| Lookahead | Sets base latency: 0, 25.3, 50.6, 76 or 100 ms; default 25.3 ms. Changing it may require the host to update delay compensation. |
| Window | Sets actual detection length from 0 to the selected Lookahead. Changes the sound while retaining the selected latency. |

> **Window shortens effective Lookahead at fixed latency**
> Compare Lookahead 50.6 ms / Window 25.3 ms with Lookahead 25.3 ms / Window 25.3 ms. With other parameters identical, settings settled and latency aligned, the dynamics core produces the same compression and harmonic coloration. The first setting simply retains 25.3 ms more base latency. This lets you audition a shorter effective Lookahead without repeatedly changing plugin latency.

## Example: keep Lookahead at 25.3 ms

| Window | Actual detection length | Base latency |
| --- | --- | --- |
| 25.3 ms | Full 25.3 ms | 25.3 ms, unchanged |
| 10 ms | Reduced to 10 ms | 25.3 ms, unchanged |
| 0 ms | Current-sample detection | 25.3 ms, unchanged |

Base latency excludes OS filters and Limiter Ceiling latency. With other settings fixed, adjusting only Window leaves the reported total latency unchanged too.

## Why this helps continuous listening comparisons

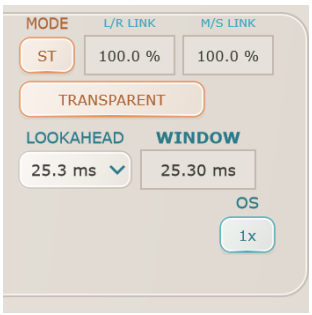
Changing Lookahead directly may make the host adjust delay compensation, causing a brief pause. Window retains the same latency, allowing continuous comparison of attacks, tails, dynamics and coloration without that compensation change.

## Window is a detection duration, not an update interval

Window = 10 ms does not mean compression happens once every 10 ms. Each gain calculation considers peaks within 10 ms before and after the current sample; Normal / Transparent then chooses between them. Detection moves and gain updates on every sample. Shortening Window changes the reference range, not the update rate.

# Lookahead and Window: controls and 0 ms

Choose the sound with Window first, then decide whether to reduce Lookahead latency.




Lookahead and Window share a row in the lower-right controls. Window always uses milliseconds.

| Lookahead | Window range | Default / Alt-click |
|---|---|---|
| 0 ms | Fixed at 0 ms; not editable | 0 ms |
| 25.3 ms | 0-25.3 ms | 25.3 ms |
| 50.6 / 76 / 100 ms | 0 to selected Lookahead | Equals selected Lookahead. |

| Adjustment | Result |
|---|---|
| Double-click / vertical drag / Shift-drag | Type milliseconds / continuous adjustment / fine adjustment. |
| Alt-click Window | Restore the maximum allowed by the current Lookahead. |
| Change Lookahead manually | Window follows the new Lookahead and resets to its maximum. |
| A/B, project restore, reopen editor | Recall saved Window; do not treat recall as a manual reset. |

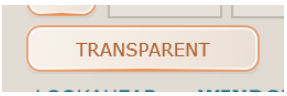
> **25.3 ms / Window 0 ms versus 0 ms / Window 0 ms**
> With the same algorithm, parameters and OS, both detect the current sample. The first retains 25.3 ms of base latency; the second removes it. If Window = 0 ms sounds best, try 0 ms Lookahead, allowing the host to finish delay compensation. For Limiter listening comparisons, see page 39.

Most normal compression can retain the full Window. Shortening it can increase local dynamics, harmonic coloration and crackles. At low values, especially below 10 ms, oversampling is recommended; see page 45. Window avoids pauses caused by changing delay compensation; it still changes the processing sound.

Old 10 / 26 ms Lookahead settings migrate to 25.3 ms; 40 ms to 50.6 ms; 80 ms to 76 ms; and 105 ms to 100 ms. Window remains adjustable within the current range. A 0 ms Lookahead setting does not guarantee zero total latency: OS filters and Limiter Ceiling can still add delay.

# Transparent: detail and detection

Choose a detection strategy without changing the selected Lookahead latency.




Transparent is on the second row at the lower right. Yellow means Transparent; unlit means Normal. New instances default to OFF.

Transparent (formerly SAFE) lets you compare transients, detail and detection stability, and can help reduce crackles. Set Lookahead and Window first: Window sets the length; Transparent chooses how to use its past and future peaks.

| Detector | How past/future peaks are used |
|---|---|
| Off (Normal) | Find the peak in each past/future Window, then use the smaller one. Usually retains more local dynamics and reduces a loud event's influence on nearby quiet audio. |
| On (Transparent) | Use the larger peak. A longer Window stabilizes detection and helps suppress continuous crackles caused by rapid gain changes. |

## When to try Transparent

Compare attacks, detail and sustain with Transparent on and off. For continuous crackles in clean guitar, vocals with delay / reverb, or Single mode with Range enabled, try Transparent with the full Window first.

> **The tradeoff: pre- and post-attenuation**
> An upcoming loud event can attenuate nearby quiet audio before it arrives. After it ends, the past Window may still contain its peak and attenuate following quiet audio. Dips may appear on either side of loud events. A shorter Window reduces their duration, but also weakens crackle suppression.

## How Transparent differs from Classic / Super

Transparent changes how the detector level is selected. Classic / Super chooses the dynamics curve applied to that level. Pre- and post-attenuation come from the selected Window; no hidden Attack / Release stage is added to the dynamics core.

## Default state

Transparent defaults off.

## Turning Transparent on and off

Click Transparent to turn it on (yellow); click again to turn it off (unlit). Window and latency stay unchanged. The on/off state is saved with the project and A/B.

> **Window 0 ms and Lookahead 0 ms**
> With Lookahead 25.3 ms / Window 0 ms, detection uses the current sample but retains 25.3 ms of lookahead latency. Lookahead 0 ms removes that delay and fixes Window at 0. Both cases give identical Normal/Transparent detection: Transparent is disabled and greyed out, retaining its on/off choice. It becomes available when both values exceed 0 ms. OS and Limiter Ceiling may still add latency.

# Transparent: transients, detail and crackles

Compare attacks and detail first, then listen for crackles and changes around loud events.

1. **Start with a full-Window comparison**
   Start at Lookahead 25.3 ms, Transparent off and Window 25.3 ms. Fix Ratio, thresholds, Makeup, Mix and Output, and listen for crackles and the main sound's dynamics.

2. **Enable Transparent with the full Window**
   Transparent uses the larger past/future peak. A longer Window helps suppress crackles caused by rapid gain changes. Compare attacks, sustain and repeated echoes with Normal, not just loudness.

3. **Shorten Window gradually**
   If audio around loud events is attenuated too much, shorten Window gradually to balance crackle control against pre-attenuation. Lookahead and base latency stay unchanged. Alt-click restores the full Window.

4. **Compare overall OS last**
   At the selected Window, compare 1x, 4x, 8x and 16x. Higher OS reduces aliasing but does not replace Window selection. Finish with a level-matched A/B comparison and check final output peaks.

Range in normal Single mode can also cause crackles on some material. Try Transparent here too, starting with the full Window and shortening it by ear.

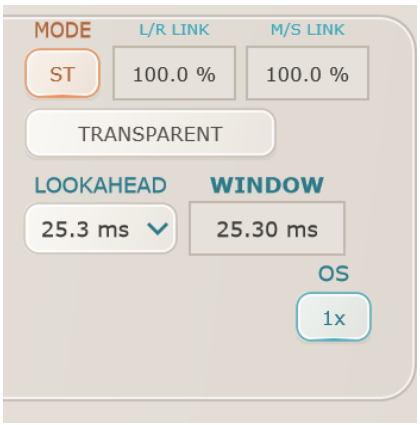
| Goal | Try first |
| --- | --- |
| Retain more local dynamics | Start with Transparent off and the full Window. |
| Add coloration deliberately | Shorten Window gradually in Normal, listen for crackles and roughness, and enable OS. |
| Reduce Window-related crackles | Enable Transparent with the full Window, then shorten it to suit the source. |
| Shorten Transparent's surrounding dips | Shorten Window; protection also weakens. At Window 0 ms, both detectors are identical. |

> **Listen to both sides of the Transparent tradeoff**
> A longer Window can attenuate quiet audio before a loud event arrives and keep attenuating briefly after it ends. This is pre- and post-attenuation, not an added Attack / Release stage. Transparent targets this type of rapid gain modulation; it cannot repair clipped input or host dropouts.

No visible dip does not mean equal distortion at two Window settings: gain may vary differently within steady cycles. Listen to actual audio and examine harmonics and level steps together. The plug-in Display helps explain settings but does not replace these checks.

# Two ST Links, one set of parameters

Understand left/right and mid/side, then adjust their coupling.




Comp / ST: L/R Link and M/S Link appear together beside Mode. Both default to 100%.

L/R means left and right channels; M/S means mid and side signals. Mid reflects what the channels share, while Side reflects their difference. They are two ways of describing the same stereo material.

## Link controls how closely compression is coupled

| Control | As the value decreases |
|---|---|
| L/R Link | Blends in independently compressed left/right results, giving each channel's own dynamics more influence. |
| M/S Link | Blends in independently compressed mid/side results, giving the center and sides' dynamics more influence. |

At 100% / 100%, both sides share compression gain, preserving the original ST behavior. Lowering Link still uses one set of Ratio, threshold, Makeup and Mix controls; no second set of knobs appears.

## A first adjustment

1. **Start at 100% / 100%**
   Set the basic sound with Ratio, Threshold and Mix.

2. **Adjust one Link at a time**
   Leave the other at 100%. Gradually lower L/R Link or M/S Link and compare attacks, tails, the center and width.

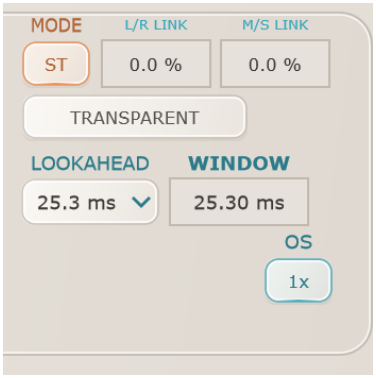
3. **Then try combining them**
   Once you can hear each change clearly, adjust both. The next page gives four starting points.

Double-click to enter a percentage; drag vertically to adjust, with Shift for finer changes. Alt-click (Option on Mac) resets to 100%.

# ST Link: four starting points

These values blend processing results; they are not channel volume controls.

| L/R Link | M/S Link | Processing result |
|---|---|---|
| 100% | 100% | Original ST behavior, with shared compression gain. |
| 0% | 100% | Independent left/right compression, using the same parameters for both channels. |
| 100% | 0% | Independent mid/side compression, using the same parameters for mid and side. |
| 0% | 0% | Equal parts independent left/right and independent mid/side processing. |




Both Links at 0%: combine independent L/R and M/S processing with one set of controls.

## Use intermediate values to fine-tune

With M/S Link at 100%, lowering L/R Link gradually moves from shared gain toward independent left/right compression. Lowering both blends in independent left/right and independent mid/side results together.

The values are not volume percentages and do not need to add up to 100%. This is not L/R compression followed by another M/S compressor in series. The two values determine how the processing results are combined.

> **Numeric Link and parameter LINK are different**
> These two values change the coupling of audio processing. In LR / MS, the LINK button beside Mode moves two sets of parameters together; it does not force both sides to use identical compression gain.

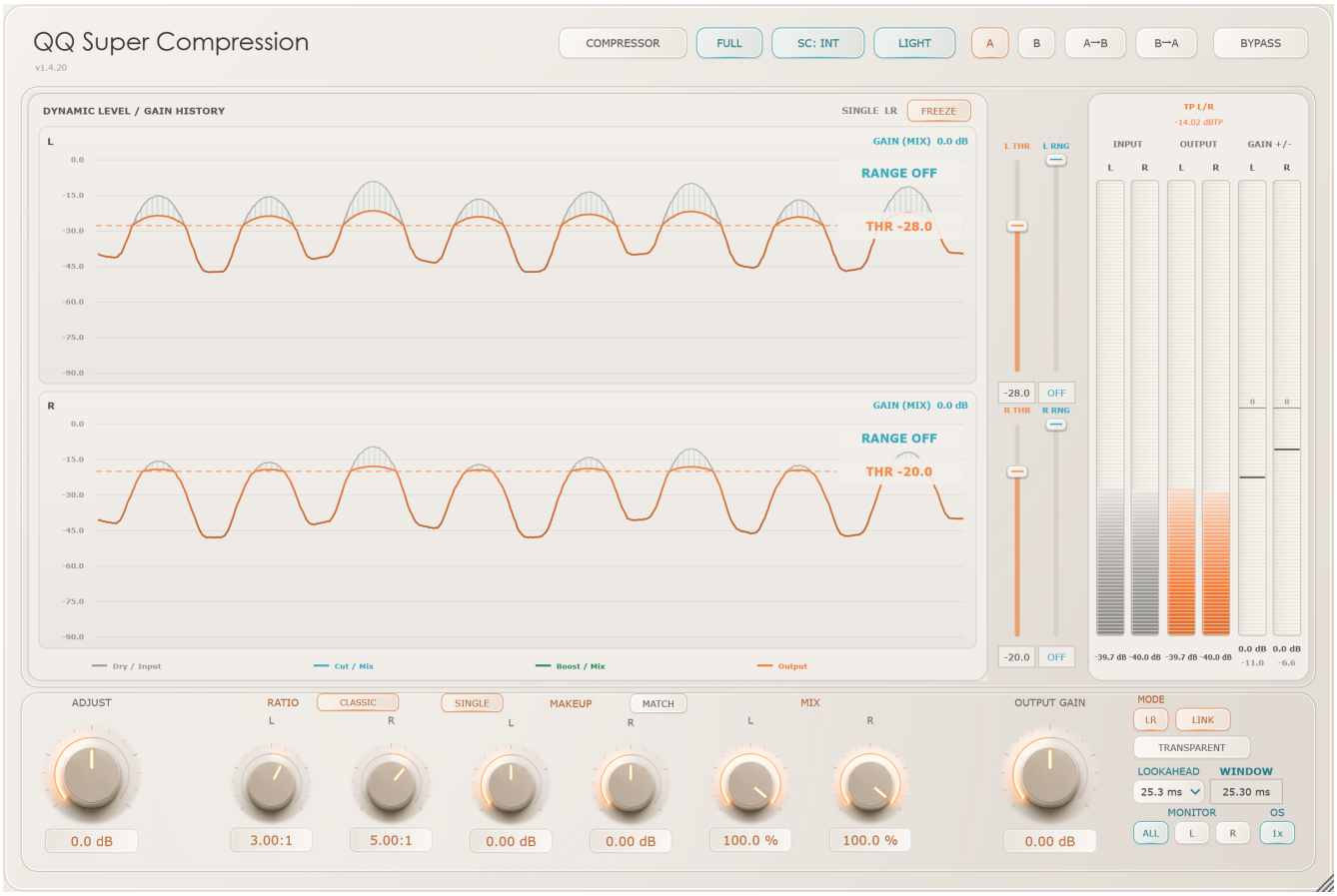
## When should you switch to LR / MS?

Choose LR / MS when left/right or mid/side need different Ratio, threshold, Makeup or Mix settings. Stay in ST if you only want to adjust coupling while keeping one shared set of parameters.

Lowering Link can change the stereo image and width. Use Match for a similar-loudness comparison, then check the full mix. Neither 100% nor 0% is a fixed answer for every source.

# ST and LR: shared or separate controls

In Compressor, the lower-right Mode button cycles ST → LR → MS → ST. Start with ST; choose LR when you need separate settings for each side. Clipper and Limiter omit this selector and use one ST-style graph, one set of parameters and a numeric L/R Link. All three Mode buttons share the same size; LR / MS place parameter LINK directly to the right.

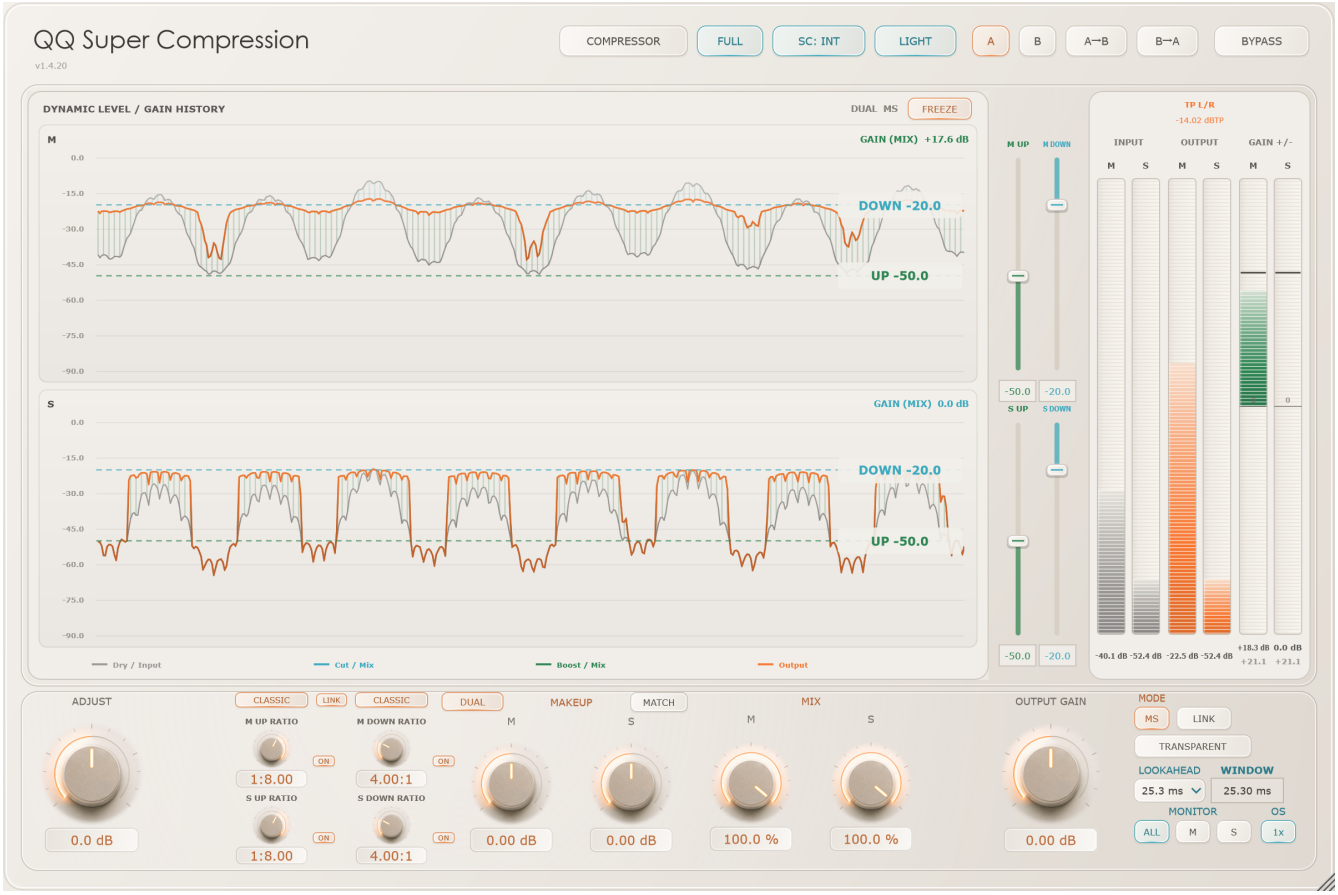



LR: L above, R below, with independent controls.

| Mode | How to use it |
|---|---|
| ST | One set of parameters. With both numeric Links at 100%, compression gain is shared. Lower them to blend in independent left/right or mid/side compression. |
| LR | Left and right have independent ratios, thresholds, Makeup and Mix; DUAL adds UP / DOWN per channel. Check for stereo image shifts on asymmetric material. |

**MONITOR**
Choose L or R to audition that side centered. Return to ALL to check normal stereo, and do not leave one-sided audition active for the final output.

# MS: control center and sides




MS / DUAL: Mid is above Side. Each channel has its own UP / DOWN Ratios and thresholds.

Mid mainly represents content common to both sides; Side represents their differences. They are not a precise separation of a center track from every other track.

1. **Listen to M and S first**
   Use M / S under MONITOR, then return to ALL.

2. **Adjust each for a reason**
   For steadier center dynamics, start with M Ratio / Mix. Treat S cautiously when preserving ambience and width.

> **Recheck the full mix**
> Compression or level changes in M/S can alter the sense of space. In ALL, check center, width and mono compatibility, not only soloed domains.

# Channel LINK: adjust together, keep differences

In normal LR / MS, LINK beside Mode connects matching controls across channels. Set the channels separately, then enable it to adjust them together without forcing equal values.




SINGLE example: channel LINK beside Mode is on, while the left and right Ratios remain at 3:1 and 5:1.

| Parameter | Before | After an equal change |
|---|---|---|
| Ratio | 3:1 / 5:1 | 4:1 / 6:1 |
| Threshold | -20 / -10 dB | -18 / -8 dB |
| Makeup | -3 / +1 dB | -2 / +2 dB |
| Mix | 100% / 70% | 90% / 60% |

1. **Set the relationship with LINK off**
   Decide which domain needs more compression or a different blend.

2. **Enable LINK for joint adjustments**
   Adjust either channel's control and the other follows by the same amount. In Dual, toggling UP ON synchronizes the other channel's UP ON; DOWN ON works the same way. Upward and downward switches remain independent.

> **Linked controls stop together at their limits**
> If either side would exceed its range, both stop where their difference can be retained. For example, thresholds of +10 / +20 dB can rise together only as far as +14 / +24 dB. Classic and Super follow the same rule.

## Numeric Link values and LINK buttons

ST L/R Link and M/S Link control audio coupling; the LINK button beside Mode moves two sets of parameters together. The small LINK between the Ratios couples Up / Down inversely. Mode LINK couples matching L/R or M/S parameters. With both buttons on, channels move together while each channel retains its Up / Down relationship. Clipper / Limiter Output LINK connects the downward threshold, Makeup and Output; see page 44.

# Sidechain: start with the panel

Click SC: INT / SC: EXT at the top right. Its label shows the current source.




| Control | How to use it |
|---|---|
| INT / EXT | INT follows the current track itself. EXT uses a signal sent from another track to control boost or attenuation on the current track. |
| KEY GAIN | Available only with EXT. Adjusts the external detector level and trigger strength, not the main track level. |
| HPF | Reduces low frequencies in the INT or EXT detector key. OFF is full-band; active range is 20-500 Hz. |
| KEY LEVEL | Check that the detector receives a signal. With EXT showing N/A, first check that the host sidechain bus is enabled. |
| SC LISTEN | Temporarily audition the detector key to check routing and HPF. Switch it off before judging normal output. |

**Two common tasks**
If bass content compresses everything, try INT + HPF; see page 27.
To make bass duck when the kick arrives, use EXT; see page 28.

The EXT curve shows the external key after HPF and Key Gain. Window, Transparent and OS do not change this curve; HPF and Key Gain do. It is time-aligned with the main audio history.

# Internal sidechain: reduce bass dominance

Use this when kick or bass repeatedly pulls down the rest of a drum bus, mix bus or master. The goal is less low-frequency triggering, not less bass in the audible signal.

1. **Start with INT and hear the problem**
   Loop a passage with bass and other content together. Leave HPF OFF to establish a comparison.

2. **Open the sidechain panel**
   Click SC: INT and keep SOURCE on INT. A disabled Key Gain is normal here.

3. **Raise HPF gradually**
   Start at a low cutoff and listen for less whole-mix ducking on bass events. Make small changes.

4. **Audition the key if needed**
   Use SC LISTEN to hear which bass content HPF removes, then turn it off and return to the full sound.

5. **Inspect history while dragging**
   HPF history also refreshes during dragging and finishes after release. If HPF UPDATING appears, wait for it to clear before a precise comparison.

6. **Rebalance the compression amount**
   Filtering may reduce GR or change its shape. Revisit Ratio, Threshold or Mix as needed, then compare at similar loudness.

> **It is normal not to hear bass being cut**
> HPF filters only the key used to decide compression, not normal output. It is not equivalent to Key Gain: Key Gain scales the detector level; HPF changes frequency weighting. Too high a cutoff can make the compressor ignore bass that still needs control.

# External sidechain: kick-controlled bass

Insert the plugin on the bass - the track to turn down. Send the kick to its sidechain as the trigger. Do not reverse the tracks or confuse a normal audio send with a sidechain send.

KICK TRACK → BASS PLUG-IN KEY → BASS IS REDUCED

1. **Enable the host sidechain input**
   Enable sidechain input for this plugin instance on the bass. Add and enable a kick send to that sidechain. Names and locations vary by host.

2. **Select EXT in the plugin**
   Open SC, choose EXT and play the kick. Check KEY LEVEL first; if it has no reading, fix routing before increasing Ratio.

3. **Confirm the actual trigger**
   Briefly enable SC LISTEN: you should hear the kick, not the bass. Turn audition off afterward.

4. **Set trigger level and reduction**
   For downward ducking, start with SINGLE, Ratio above 1:1 and Range OFF. Set detector strength with Key Gain or the send level, and set the lower boundary with Threshold. Leave HPF OFF initially.

5. **Use Mix to decide how much room to make**
   Find an audible duck, then reduce Mix to retain the desired bass sustain. Listen to both tracks together, not only solo bass.

6. **Decide whether HPF helps**
   Raising HPF reduces low-frequency kick triggering and may weaken the duck you want. Use it when you want more emphasis on the attack or less unwanted low-bass triggering.

There are no conventional Attack / Release controls. Ducking depends on the trigger material, Lookahead, Transparent, Window and compression settings. It is not necessarily smoother than conventional sidechain compression on every source.

# Apply it in a real mix

## Vocals, guitar, piano: consistency with articulation

Try SINGLE / ST / INT and 25.3 ms. Raise Ratio above 1:1, then use Mix to keep consonants, plucks or hammer attacks natural. To process only louder parts, raise Threshold and leave Range OFF. Use Match and compare in the mix.

If clean guitar or vocals containing delay / reverb produce continuous crackles, enable Transparent with the full Window, then shorten Window gradually and compare; see page 20.

## Tails and quiet phrases: bring detail forward

To lift quiet syllables, plucked-note tails or ambience, try upward Single compression. Use Threshold to exclude tiny sounds you do not want to lift, and Range to exclude louder parts. To control loud parts too, switch to Dual and adjust Up and Down separately. Use their independent ON / OFF switches to compare; disable Ratio LINK for independent ratio adjustments.

## Mix bus / mastering: a more cohesive balance

When you want a steadier, more cohesive mix without deliberately reshaping drum attacks through conventional Attack / Release, try it as an alternative to G Bus-style compression. Start gently, compare balance across sections and lower Mix when needed.

If every kick pulls the whole mix down, try INT + HPF. Do not raise the cutoff merely to make the GR number smaller. At similar loudness, check drum attack, low end, space and the song's larger dynamic movement.

## Vocal-keyed accompaniment: make space for words

Insert the plugin on the backing bus, send the vocal to its external sidechain and choose EXT. Confirm the source with SC LISTEN. Select SINGLE, Ratio above 1:1 and Range OFF, then adjust Key Gain, Threshold and Mix so the backing steps back while the vocal plays.

If plosives trigger too much ducking, try raising sidechain HPF. If breaths or noise also make the accompaniment dip, check the sent signal and Threshold; HPF alone will not solve every unwanted trigger.

> **For every comparison**
> Turn off SC LISTEN. In normal LR / MS, return MONITOR to ALL. Check that before/after loudness is similar. For final peak control, enable Limiter and set Ceiling; use headphones and Match for level-matched listening, as on pages 33-39.

# Clipper: strong downward compression

One set of parameters and L/R Link, without Ceiling.




CLIPPER has no dual compression or Ceiling. LINK beside Output defaults off.

Clipper uses the Classic / Super dynamics algorithms for downward compression only. Ratio ranges from 10:1 to 1000:1, letting you compare stronger compression's effect on peaks, density and tone. Lookahead, Window, Transparent, Mix and OS remain available.

| Control | Behavior |
|---|---|
| Ratio | 10:1-1000:1; default and Alt reset: 10:1. The arc is unlit at the far left to show the minimum setting. |
| L/R Link | 0%-100%; default and Alt reset: 0%. Left and right can act independently; increase the value for more coupling. |
| Output LINK | Off by default: threshold, Makeup and Output adjust independently. Enable it for equal and opposite linked changes. |
| Simplified controls | Downward only. No dual mode, Range, Ceiling, channel Monitor or ST / LR / MS selector. LUFS-I is hidden; Match remains available. |

> **How Clipper differs from Limiter**
> Clipper has no final Ceiling protection. Use Limiter's Ceiling when you need a final peak limit. Clipper's threshold readout is not an output ceiling after subsequent gain adjustments.

# Clipper: Window, coupling and GR hold

Listen first; use GR to see the deepest reduction during this playback.

1. **Start with gentler downward compression**
   Choose Classic or Super, start Ratio at 10:1 and gradually lower Threshold. Keep Output LINK off; adjust Makeup or Output independently as needed for a level comparison.

2. **Adjust Window at fixed latency**
   Keep Lookahead at 25.3 ms and shorten Window to compare peaks, density and coloration. Transparent is greyed out at Window 0. If that sound works best, consider switching to 0 ms Lookahead.

3. **Adjust how the channels influence each other**
   Start L/R Link at its default 0% for independent left/right compression. Increase it for more shared attenuation, listening to the image and dynamics as you adjust.

4. **Compare OS and final loudness**
   For short Window settings or deep compression, compare 4x / 8x / 16x. OS defaults to 1x and is your choice. Finish with a similar-loudness comparison using Match, Bypass or A/B.

## Gain Reduction: live bars and held maxima

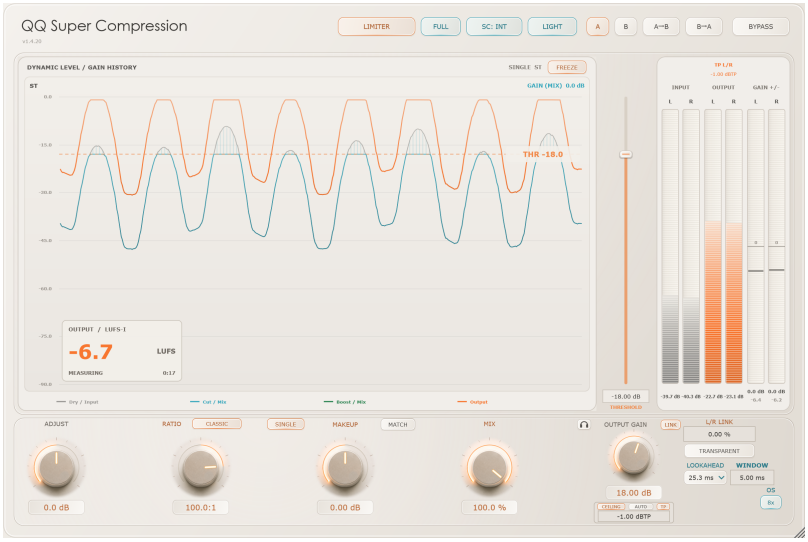
The left and right Clipper bars show current gain reduction in real time. Each number and marker holds the deepest reduction reached during this playback, updating only when reduction goes deeper. Held values remain after stopping. Makeup and Output Gain are excluded.

| Adjustment | GR history |
| --- | --- |
| Stop playback | Bars still show current reduction; numbers and markers retain the maximum. |
| Double-click the Gain Reduction area | Clears both channels' GR and starts accumulating again. |
| Start playback again | Resets automatically and records this new playback. |

> **Clear GR and TP separately**
> Double-clicking GR clears only the reduction history. Double-click the top TP number to clear the true-peak hold. Maximum GR does not mean that amount is applied continuously; listen to the changing dynamics too.

# Limiter L/R Link: connect left and right limiting

One ST Display and one set of controls, with independent left/right limiting.




The first row at the lower right is numeric L/R Link. Limiter has no ST / LR / MS selector or channel Monitor.

| L/R Link | Response |
|---|---|
| 0% (default / Alt reset) | Each channel detects and controls its own attenuation. A loud event on one side does not attenuate the other merely through linking. |
| Between 0% and 100% | Gradually increase shared attenuation; choose the balance between independent dynamics and stereo stability by ear. |
| 100% | Apply the stronger attenuation to both channels to reduce image movement caused by different gain changes. |

L/R Link applies to dynamics and final Ceiling. Upward gain is not transferred to the other channel merely by linking. Double-click to type a percentage, drag vertically, or hold Shift for fine adjustment. Alt-click resets to 0%.

## Loudness, detail and stereo stability

Start at 0% for maximum loudness. Increase L/R Link when stereo detail and image stability matter more. The channels then share more attenuation; this does not restore detail already lost to compression. Compare at similar loudness.

## Shared settings do not mean identical detection

Ratio, Makeup, Mix and thresholds have one shared set of controls. Both channels use these settings. Dual UP / DOWN ratios remain independent, with no Ratio LINK. LINK beside Output still connects the downward threshold, Makeup and output compensation; it is separate from the audio linking amount here.

> **Opening older Limiter sessions**
> Old ST loads with 100% L/R Link; old LR / MS with 0%. Saved thresholds are restored within the current -inf to +24 dB range. Old Range no longer applies. Values previously set to -45 dB cannot be recovered automatically. The old L settings (M for MS) become shared controls; old MS now processes left/right. Different old channel settings may change the sound, so check by ear. Compressor LR / MS is unchanged.

# Limiter: compression, Link and Window

Set downward compression first, then balance loudness, dynamics and stereo detail.




One ST graph and shared parameters. Drag a rectangle to zoom into detail; threshold scales follow the graph. Limiter does not use Range.

1. **Start with downward SINGLE compression**
   Begin with Ratio at 20:1 or more, up to 1000:1, and Mix at 100%. Set Ceiling and TP; keep LINK beside Output enabled. Lower Threshold gradually: Output compensates by the same number of dB in the opposite direction.

2. **Choose the stereo relationship with L/R Link**
   For greater loudness, start at the default 0% for independent limiting. Increase it for greater stereo detail and image stability. More attenuation is shared, which can also affect loudness.

3. **Balance loudness and dynamics with Window**
   Normal compression can usually retain a full Window. In Limiter, Window is especially important for loudness and dynamics. Keep Lookahead fixed and shorten Window gradually. At low values, especially below 10 ms, use oversampling.

4. **If Window 0 sounds best, try 0 ms Lookahead**
   With other parameters and OS equal, both use Window = 0 ms detection. Choose the sound at the existing latency first, then select 0 ms to reduce base latency. Allow the host to update compensation when changing Lookahead; see page 39.

> **Distinguish the two links**
> L/R Link controls shared left/right attenuation. Output LINK connects the downward threshold, Makeup and output compensation. After these adjustments, consider adding a little UP, then compare at matched loudness with headphones, Match and Bypass.

# Limiter Dual: add gentle upward compression

Use Dual to add compensation after setting the standard downward compression.




Example: DOWN remains 100:1, UP is 1:1.2 and Output LINK is on. Up and Down ratios are independent; there is no Ratio LINK.

1. **Complete the previous page's downward setup**
   Use a Ratio of 20:1-1000:1, Threshold and Output LINK to set the desired Limiter compression. Note the algorithm and Ratio.

2. **Switch to DUAL and retain the downward setup**
   Single Threshold carries over to the DOWN threshold. DOWN RATIO and Single Ratio are stored separately: check the DOWN algorithm and Ratio, and set them to your Single values if needed to retain the compression you established.

3. **Add only a little UP**
   Keep Output LINK enabled. Limiter has no Ratio LINK, so adjusting UP does not change the DOWN ratio. Enable UP and move gently from 1:1 toward 1:1.2; this small range is often enough. Use the UP threshold to choose the audio to lift.

UP remains independent of Output Gain and Makeup, even with Output LINK on. Drag a rectangle to zoom Display; right-click to restore the full view. UP cannot exceed DOWN, and zoom does not rewrite thresholds.

> **Listen closely to kick and snare sustain**
> Upward compression can raise kick and snare tails, making them more prominent. Listen beyond attacks and loudness: check whether sustain fills the spaces between hits or makes the rhythm feel sluggish.

## If Dual does not work well, try segmented compression

Use gentle upward compression to add detail to the established downward setup. Keep the original DOWN RATIO, then fine-tune if needed. If adding UP does not improve the result, reduce or disable it and try segmented compression to control dynamics; this can work better. Decide whether to use Mix after settling these choices.

# Ceiling: enable final peak protection

Available in Limiter, below Output Gain.




CEILING lights when protection is enabled. TP selects true-peak limiting.

## Ceiling sets the final peak limit

Ceiling controls peaks after dynamics, Makeup, Mix and Output Gain. Its range is -24 to 0 dB, with 0 dB as the default. CEILING is on by default in Limiter.

## Use TP to choose which peaks to limit

| State | How it works |
|---|---|
| TP on | Limits reconstructed true peaks, accounting for peaks that may occur between samples. |
| TP off | Limits ordinary sample peaks, using gain attenuation and recovery. |
| Available in both states | AUTO / TIGHT / SMOOTH recovery and the selected overall OS: 1x / 4x / 8x / 16x. |

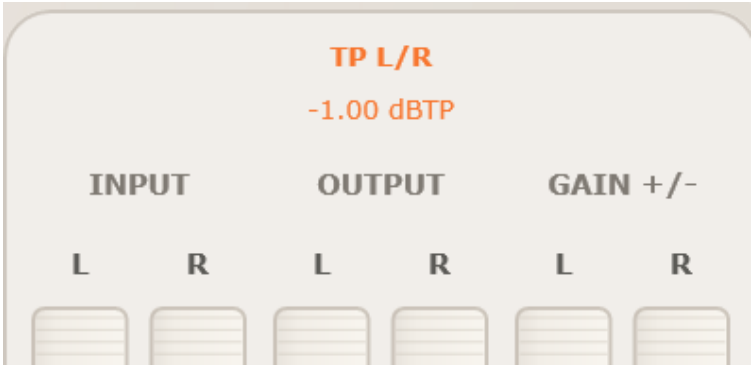
TP retains a small protection margin, so readings need not sit exactly at Ceiling. With TP off, intersample true peaks can still exceed the setting; check the TP readout at the right.

## Adjusting the value

Double-click to type, drag vertically, or hold Shift for fine adjustment. Alt-click resets to 0 dB. See page 46 for recovery modes. Toggling TP does not change the current Ceiling latency.

> **When CEILING is off**
> Final peak limiting and its additional latency are removed. Ceiling level, TP and recovery controls become inactive, retaining their settings for re-enabling. Compression, OS, Output Gain and Output LINK keep their current behavior.

# TP meter: measure output and hold peaks




The TP L/R readout measures true peaks in the final left/right stereo output.

## Measured after Ceiling

The TP meter reads actual output, including Makeup, Mix, Output Gain, enabled Ceiling and headphone 1:1 compensation. It does not simply display the Ceiling setting, or clamp its reading to that setting.

## Hold and reset

1. **Play the passage to check**
   The peak reading holds for about 20 seconds. A new higher peak updates the value and restarts the timer. Hold advances with audio playback time.

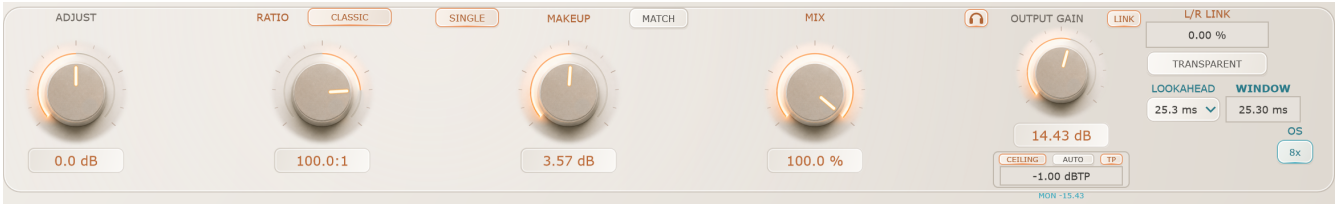
2. **Start a fresh measurement**
   Double-click the TP readout to clear it immediately, then play the passage again. There is no need to wait for the previous held peak to expire.

3. **Measure again after changing Ceiling**
   A held peak may come from earlier settings. Clear it before deciding whether the current limit behaves as expected.

| What you see | How it works |
| --- | --- |
| TP exceeds Ceiling with TP off | The meter still measures true peaks. Sample-peak protection does not also guarantee the intersample peak limit. |
| TP changes with headphones on | The readout includes listening compensation. Judge it against the converted MON limit. |
| Normal MS domain meters differ from TP | TP measures final L/R. Individual M/S levels and left/right output limits are different quantities. |

# Headphone 1:1: keep limiting, offset gain

The small headphone button left of Output Gain appears only in Limiter.




With headphones on, Output values and links stay active. MON shows the listening limit below.

With headphones on, audio still passes through dynamics and any enabled Ceiling. The visible Output Gain is then cancelled. Limiting's dynamic and waveform changes remain; only the listening level is adjusted for comparison.

## The Output value still matters

Output remains adjustable. Links between the downward threshold, Makeup and Output remain active; the Dual UP threshold stays independent. The headphone function cancels only Output's direct level change, so changes to dynamics remain audible. Switch it off to restore normal Output Gain.

## Ceiling follows the same listening reference

> **MON = Ceiling - Output Gain**
> For example, Ceiling = -1 dB and Output = +6 dB give a listening limit of -7 dB with headphones on. The Ceiling setting stays at -1 dB. Switching headphones off restores the normal output reference. This listening limit applies only while CEILING is on.

Headphones default to off, are saved with the project, and remain independent of A/B. Keep the same listening mode while comparing A/B. Headphone and Bypass switches use smooth transitions; headphones add no separate audio delay.

> **1:1 does not automatically mean equal loudness**
> Headphones cancel Output Gain. Match can compensate dynamic compression, but does not measure the final Ceiling or headphone result. Trim Makeup if needed; see the next page.

# Limiter: headphones, Match and Bypass

1. **Compare after setting up the Limiter**
   Complete downward compression on page 33, or add gentle upward compression on page 34, then enable headphones. Keep Ceiling and TP settings unchanged and turn SC LISTEN off. Limiter outputs full stereo.

2. **Play the same representative passage**
   Start playback and wait for valid measurements before clicking Match. It compares original input and compressed Wet loudness. To measure only new settings, stop and replay.

3. **Click Match to compensate Makeup**
   Match writes one shared Makeup value. Headphones, Output and Ceiling are outside this measurement; Match does not automatically compensate final loudness changes caused by them.

4. **Compare the original with plugin Bypass**
   Bypass returns delay-aligned original input without limiting or headphone gain compensation. Listen for attack, tails, density and groove; trim Makeup if needed.

5. **Finish listening and restore normal output**
   Switch headphones off and keep the adjusted settings. Check TP and Ceiling again before judging the final signal sent downstream.

## Match applies one correction, not continuous gain riding

Every click measures from the start of this playback, without clearing history or stacking the previous correction. To compare only new settings, stop and replay before Match. With Output LINK on, Match-induced Makeup changes follow the link rules. Compensation may be limited at the Makeup boundary or with a fully dry Mix.

> **When comparing A/B**
> Play the same material for A and B, then Match each. Headphone state is independent of A/B; Limiter / TP / Ceiling, rates and recovery are recalled with A/B. Keep Lookahead and rates alike where possible to reduce latency changes.

# Limiter: compare Window, then choose 0 ms

Window changes detection length; Lookahead sets base latency.

## Keep Lookahead fixed while comparing Window

After setting downward compression and L/R Link, retain Lookahead 25.3 ms and shorten Window from 25.3 ms. Reported latency stays unchanged, avoiding pauses from compensation changes while you compare loudness, attacks, tails and contrasts between sections.

With Transparent off, shorter Window increases local dynamics and coloration, but may also add roughness or crackles. With Transparent on, it reduces attenuation around loud events while weakening protection. Most normal compression needs little adjustment; Limiter rewards careful comparison.

> **If Window = 0 ms sounds best**
> You can also select Lookahead 0 ms. With other parameters and OS equal, delay aligned and switching transitions excluded, both use the same sample-by-sample, zero-Window detection. The original setting retains Lookahead latency; 0 ms removes that part. Do not subtract two unaligned renders.

## Loudness and dynamics at 0 ms

Limiter at 0 ms can increase loudness while retaining contrasts between sections. Results depend on the material, algorithm and amount. Listen to both choruses and quiet sections, rather than judging loudness alone. With Window 0, Transparent and Normal detect identically.

## Use oversampling and compare Ceiling recovery

At low Window values, especially below 10 ms, and when using Limiter Ceiling, compare 4x / 8x / 16x. Oversampling reduces aliasing, not all harmonics. With Ceiling on, compare TIGHT, AUTO and SMOOTH recovery; see page 46.

> **Lookahead is not the total plug-in latency**
> Enabled Ceiling adds two approximately 3 ms lookahead stages plus reconstruction alignment; disabling it removes this delay. OS filters can also add latency. Toggling Ceiling or changing processing modes may make the host update compensation. Adjusting only Window leaves total latency unchanged.

# Reference: modes, dynamics and Link

| Parameter | Range / choices | Notes |
|---|---|---|
| Processing mode | Compressor / Clipper / Limiter | Cycles through the modes; each stores its own parameters. |
| Algorithm | Classic / Super | Comp / Limiter Dual: separate UP / DOWN algorithms. |
| Single Ratio | Comp 1:200 to 200:1; Limiter 1:8 to 1000:1 | Both default and Alt-reset to 1:1. |
| Clipper Ratio | 10:1 to 1000:1 | Downward only. Default / Alt: 10:1; no lit arc at minimum. |
| Up / Down Ratio | Comp: up to 200; Limiter: UP 8, DOWN 1000 | Default 1:1; Clipper has no dual mode. |
| UP / DOWN switches | ON / OFF | Independent dual switches retain their Ratio values. |
| Threshold / UP / DOWN | -inf to +24 dB in both algorithms | Display scales follow zoom; stored values stay unchanged. |
| Range | -90 to +24 dB, or OFF | Comp Single only; no compression at or above this level. |
| Comp Mode | ST → LR → MS → ST | LINK beside LR / MS couples the two sets of parameters. |
| ST L/R Link / M/S Link | 0%-100% each | Both default and Alt-reset to 100%. |
| Clipper / Limiter L/R Link | 0%-100% | Default / Alt: 0%; one set of parameters. |
| Ratio LINK | Comp Dual: OFF / ON | Links UP / DOWN inversely; initially ON. |

**Threshold order**
Range stays at or above Threshold; DOWN stays at or above UP. Faders move together on contact; coincidence stops dynamics. Clipper / Limiter have no Range. Dual UP stays independent of Output / Makeup linking.

| Threshold control | New instance default | Alt / Option-click |
|---|---|---|
| Comp Single Threshold | -90 dB | -90 dB |
| Dual UP / DOWN | -90 dB / 0 dB | -90 dB / 0 dB |
| Clipper / Limiter Single Threshold | 0 dB | -90 dB |

Classic and Super share ranges and resets. LINK respects shared limits. For upward gain with Range OFF, see page 9.

**The special -inf setting**
-inf removes the finite lower threshold. Classic Single with Range OFF, Mix 100% and Ratio above 1:1 becomes silent; 1:1 passes through. Super keeps its own curve. Defaults remain -90 / 0 dB.

Recheck old threshold / Range automation: the wider range may change values at saved automation positions. Ordinary saved parameters retain their dB values.

# Quick reference: display, gain and sidechain

| Parameter | Range / choices | Notes |
|---|---|---|
| Adjust | -90 to +90 dB; default / Alt: 0 | Moves only the graph; project-saved, independent of A/B and processing mode. |
| Makeup | Comp ±30 dB; Clipper / Limiter ±120 dB | Default 0; between Wet and Mix. |
| Mix | 0 to 100% | Default 100%; blends the dynamics processing. |
| Output Gain | Comp ±24 dB; Clipper / Limiter ±120 dB | Headphones cancel the direct monitoring gain from Limiter Output. |
| MONITOR | ALL / L / R or M / S | Available only in Comp LR / MS. |
| Lookahead | 0 / 25.3 / 50.6 / 76 / 100 ms | Initially 25.3 ms; manual changes reset to a full Window. |
| Window | 0 to current Lookahead, in ms | Default / Alt: maximum. Adjust detection length at fixed latency. |
| Transparent | Default OFF | Disabled at Lookahead or Window 0 ms. |
| Overall OS | 1x / 4x / 8x / 16x | Active at every Lookahead setting; default 1x. |
| SOURCE / KEY GAIN | INT / EXT; ±24 dB | KEY GAIN is adjustable only in EXT. |
| HPF / SC LISTEN | OFF / 20-500 Hz; OFF / ON | Filters detection only. Listening ends when the panel or editor closes. |
| Processing mode / TP | Default Compressor; Limiter TP defaults ON | TP and Ceiling are available only in Limiter. |
| Ceiling | Initially ON; -24 to 0 dB | Default 0 dB. Off removes final protection and its added latency. |
| Ceiling recovery | TIGHT / AUTO / SMOOTH | Default AUTO; available with TP on or off. |
| Headphone 1:1 | Default OFF | MON = Ceiling - Output; saved in projects, independent of A/B. |
| FULL / ECO | New instances recall the last manual choice | Per instance; saved in projects, independent of A/B. |
| FREEZE | Initially ON; new instances recall the last choice | Freezes only after Stop. Per instance; saved in projects, independent of A/B. |

Each processing mode retains its gain, detector and timing settings; see page 44. ECO transport suspension, hidden-editor analysis and FULL low-residual sleep are covered on pages 48-49.

# Upward / Dual troubleshooting

| Symptom | First check |
| --- | --- |
| No audible boost after adjustment | Check that the relevant Ratio is below 1:1, Mix is above 0 and Bypass is off. In Dual, the relevant upward/downward switch must be ON. |
| No upward effect in Dual | The detector must be above UP and below DOWN. Below UP, the original level is retained; above DOWN, only downward processing applies. |
| Faders set, but no processing | Do Threshold and Range (UP and DOWN in Dual) coincide? This stops all dynamic processing. Separate them and compare. |
| Loud parts suddenly stop being processed | Is Range enabled in normal SINGLE? Compression stops at or above it. Set Range to OFF to keep processing louder audio. |
| Adjusting Up also changes Down | Normal Ratio LINK is initially enabled; disable it for independent adjustment. Limiter ratios are always independent. |
| LINK will not move further | The other control may have reached a limit, stopping both. Reverse the adjustment to continue. |
| Main track level does not explain triggering | Check whether SOURCE is EXT or HPF is changing the detector signal. Threshold decisions follow the detector signal. |

## Check Dual upward processing alone

1. **Keep other variables fixed**
   Select DUAL and turn DOWN off. In normal compression, also disable Ratio LINK. Keep Mix at 100% and all gains at 0 dB; choose a suitable Lookahead.

2. **Place the signal between the thresholds**
   For example, set UP to -50 dB and DOWN to -20 dB. Check that the detector level of a quiet passage lies mainly between them. With Ratio still at 1:1, there is no boost yet.

3. **Increase upward strength from 1:1**
   Move Up Ratio through 1:2, 1:4 and 1:8 while listening and watching Boost / Mix and positive dynamic gain. Once confirmed, enable DOWN to add attenuation and compare overall loudness again.

# Troubleshooting

## Why can Classic output fall below the threshold?

Threshold defines the gain curve at the detected level, not a floor for the final output. Quieter parts within a Window also receive gain determined by that peak. Shared gain with both ST Links at 100%, an external sidechain or HPF can also make the detected level differ from the visible signal. Makeup and Output Gain further change the output.

With Range OFF, internal detection, gains at 0 dB and Mix 100%, a steady level above Threshold remains at or above it under the Classic formula. If only the Display dips, remember that it is a reference calculated from current settings; check actual audio before attributing every dip to pre-attenuation.

| Symptom | First check |
|---|---|
| EXT has no effect | Enable the host sidechain input and send. Select EXT and check KEY LEVEL. N/A means no external bus is available. |
| Key Gain is disabled | Normal with INT: it adjusts only the external key. |
| HPF does not cut output bass | It filters the detector only. History refreshes while dragging; wait for HPF UPDATING to clear for a precise comparison. |
| Sidechain sound / narrow image | Turn off SC LISTEN; return normal LR / MS MONITOR to ALL. Limiter has no channel monitoring. |
| Match is unavailable | Play enough non-silent audio to establish valid readings. If ECO missed analysis, reopen the editor and restart playback. |
| Pause after changing Lookahead | Latency changes, so the host may readjust compensation. Also consider latency when monitoring a live recording. |
| Curves disappear below the graph | Right-click Display to restore the full view, then drag a new zoom rectangle if needed. Audio and threshold values outside the view are retained. |

See pages 32-39 for Limiter, TP, Ceiling and headphones, and pages 44-49 for advanced settings. For installation, updates and scanning, use the installation guide for your platform.

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github.com/Ziqing-Gu/QQ-Super-Compression

# Three modes, three sets of settings

Compressor, Clipper and Limiter store parameters separately.

The top button cycles Compressor → Clipper → Limiter. Each mode stores its own Makeup, Mix, Output, algorithms, Ratio, thresholds, sidechain, Lookahead, Window, Transparent and OS. Each starts with its own defaults on first use.

| Mode | Mode-specific settings |
| --- | --- |
| Compressor | ST / LR / MS, two ST Link values, Single Range, dual compression and Ratio LINK. |
| Clipper | Downward only, Ratio 10:1-1000:1 and numeric L/R Link. Output LINK defaults off. |
| Limiter | Numeric L/R Link, single / dual mode, Ceiling / TP and recovery. Output LINK defaults on. |

## A/B stores all three modes

Switching A/B restores all three sets of mode parameters in that snapshot, along with the active processing mode. Adjust, FULL / ECO, headphone 1:1 and channel monitoring are working preferences, independent of A/B.

## Single / Dual within Limiter

Switching to Dual carries Single Threshold into DOWN; returning to Single carries the current DOWN into Threshold. Each keeps its own Ratio. Makeup, Mix and Output retain their current settings. UP cannot exceed DOWN and is moved to the same position if necessary.

## Output LINK depends only on its switch

| Action with LINK on | Linked result |
| --- | --- |
| Increase downward Threshold / DOWN by 1 dB | Output falls by 1 dB; Makeup stays unchanged. |
| Increase Makeup by 1 dB | Output falls by 1 dB; threshold stays unchanged. |
| Increase Output by 1 dB | Downward threshold falls by 1 dB; Makeup stays unchanged. |

> **Linking still works at Ratio 1:1 or with DOWN off**
> Only Output LINK controls this relationship. Ratio, ON/OFF, Mix, algorithm, Window, OS and other parameters do not affect it. Decreasing values reverses the direction; both controls stop together at a range limit. Dual UP always stays independent. Clipper follows the same rules when LINK is enabled.

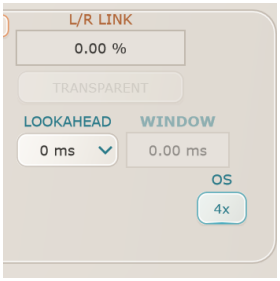
For example, lowering the downward Threshold from -18 to -20 dB raises Output by 2 dB, even at Ratio 1:1. With LINK off, the controls adjust independently.

# When to enable oversampling

One OS control covers dynamics processing and Limiter Ceiling.

## Oversampling is recommended in these three situations

| Situation | Why use OS? |
| --- | --- |
| 1 Lookahead is 0 ms | Rapid gain changes at Window 0 ms can produce more harmonics and aliasing. Compare 4x / 8x first, then increase if needed. |
| 2 Low Window, especially below 10 ms | Shorter Window can increase harmonics and modulation coloration with Transparent on or off. Use OS, then choose a rate by sound and CPU load. |
| 3 Limiter is in use | Ceiling clipping or limiting produces harmonics; OS reduces their aliasing. Overall OS works with TP on or off. |




OS is in the fourth row at the lower right. Each mode stores its own factor and starts at 1x; old projects retain saved settings.

| Rate | How to choose |
| --- | --- |
| 1x | Default for all three modes. No audio oversampling, with lower processing load. |
| 4x | A lower-load starting point. Compare roughness, high frequencies and sustained tails. |
| 8x / 16x | Higher processing rates. Choose by sound and CPU headroom; these are not enabled automatically. |

Clipper at a high Ratio and short Window also benefits from comparing OS settings. These are listening suggestions, not a sharp 'distortion-free' boundary at 10 ms. Oversampling mainly reduces aliasing; it does not remove every harmonic or replace Transparent / Window for controlling crackle and anticipatory attenuation.

> **Window retains the selected base latency**
> At Lookahead 25.3 ms, moving Window from 25.3 to 0 ms leaves base latency at 25.3 ms. The host also accounts for OS filters and Limiter protection. After changes that affect latency, let compensation settle before comparing.

Older sessions' OS setting, once used only at 0 ms, now covers all Lookahead settings. The former Ceiling OS no longer controls a separate rate. At 1x, TP can still use its independent reconstruction detector without upsampling the full audio path.

# Ceiling recovery: TIGHT / AUTO / SMOOTH

Click the small button between Ceiling and TP to choose how gain recovers after peaks.




AUTO is shown. All three recovery modes are available with TP on or off.

| Mode | Character and listening comparison |
|---|---|
| TIGHT | Releases peak attenuation faster, allowing short-term loudness to recover readily. Listen for stronger fluctuations on bass or sustained amplitude modulation. |
| AUTO | The default. Adjusts recovery to signal cycles and recurring changes, balancing isolated peaks and continuous variation. |
| SMOOTH | Slower, more gradual recovery. Compare bass stability and movement after deep limiting; attenuation may persist longer. |

1. **Keep the same passage and ceiling**
   Keep Ceiling, Ratio, Output, Mix and overall OS fixed; change only the recovery mode.

2. **Listen to short peaks and sustained bass**
   Short drum hits, sustained bass and repeated level changes can reveal different trade-offs. Do not judge from a single peak.

3. **Compare at matched loudness again**
   Recovery affects average loudness. Match does not measure Ceiling recovery; trim Makeup if needed, then compare tails, bass and attacks.

> **Same protection setting, different recovery shape**
> These choices do not change host latency at the current rate. They work with TP on or off. Disabling CEILING suspends their effect while retaining the choice. Projects and A/B save the recovery mode.

FULL near-silence sleep waits for tails and TP recovery to finish; SMOOTH may extend that wait. ECO suspends immediately and cuts tails when the host reports Stop; see page 49.

# Output LUFS-I: loudness over a passage

OUTPUT / LUFS-I appears at the lower-left of the Display in Limiter.




Accumulated output loudness, with measurement duration at the lower right. This example is not a target value.

LUFS-I measures integrated loudness over a passage; TP measures peaks. They serve different purposes. LUFS-I reads the actual monitor path, including current gain and audition settings. In Limiter, this also includes Ceiling and headphone compensation.

| State | How it works |
|---|---|
| READY | No measurement accumulated yet. Play the passage you want to assess. |
| MEASURING | Accumulating output loudness from the playback signal. |
| HOLD | Retains the last measurement after stopping. |
| --.- | No valid loudness reading yet. A signal that is too brief or too quiet may not produce one. |

A new playback, jump or loop starts a fresh measurement; stopping holds the result. If the host provides no transport state, Limiter measures continuously.

## Use the same conditions for every comparison

Play the same passage with identical SC LISTEN and headphone states. Clipper / Limiter output full stereo and have no channel MONITOR. Turn headphones and SC LISTEN off when judging the normal signal sent downstream.

> **It does not automatically set a target loudness**
> LUFS-I is a reading; Match is a one-time loudness adjustment written to Makeup. They have separate measurement purposes: a LUFS-I reading is not a Match completion signal. ECO pauses analysis with the editor hidden; reopening restarts LUFS-I. Match requires complete analysis of the current playback; restart playback if any was missed. See page 14.

Clipper hides LUFS-I but still offers Match. Loudness measurement uses gating, so very low levels may not produce valid LUFS-I. Also listen to attacks, density, tails and placement in the full mix when judging changes.

# FULL / ECO: choose for each instance

Clicking FULL / ECO at the top changes only the current plugin instance.




The ECO button. Keep an instance in FULL when it needs continuous analysis.

The table covers playback, recording and offline rendering. Host Stop behavior is described below and on page 49.

| State | Editor visible | Editor closed / hidden |
|---|---|---|
| FULL | Normal processing and analysis | Keeps background analysis and live monitoring active. |
| ECO | Normal processing and analysis | Pauses Display, ordinary meters, LUFS and Match analysis; audio processing continues. |

## Ten instances: one FULL, nine ECO

For example, choose FULL for an instance that must monitor live input while stopped, and ECO for the other nine. Instances are independent, project-saved and unaffected by A/B. New instances recall the last manual choice; saved projects keep their own choices.

## Reopening the editor in ECO

Reopening clears the old graph and starts displaying new audio, even with Freeze on. Limiter LUFS-I starts again. If the editor was closed during this playback, Match is unavailable; keep it open, stop and restart playback to measure again. Applied Makeup, Output and other settings remain.

> **ECO preserves quality and latency while processing**
> During playback, recording and offline rendering, ECO processes at the current settings. Closing the editor pauses analysis without lowering overall OS or removing compensation. On a known host Stop, it mutes and suspends processing regardless of input noise.

## At Stop, the modes differ

When the host explicitly reports Stop, ECO immediately mutes and skips compression, OS and Ceiling, regardless of editor visibility. FULL retains live input monitoring and sleeps only after silence or extremely low residual input and stable internal tails.

# Suspension and recovery after Stop

ECO follows the DAW transport; FULL retains live monitoring.

## ECO: Stop suspends processing

When the host explicitly reports Stop, with neither recording nor offline rendering active, ECO zeros the main output in that block and suspends dynamics, overall OS and Ceiling. Hardware noise or an active sidechain does not prevent suspension; remaining delay and TP tails are cut off. Editor visibility does not change this behavior.

Use FULL on an instance that must monitor live input while stopped. ECO keeps processing when the host provides no reliable transport state.

## When processing resumes

| Host state / action | Response |
| --- | --- |
| Playback or recording starts | Full processing resumes in the first callback block. |
| Offline rendering | Full processing continues, even if Play is not flagged. |
| Switch to FULL while stopped | Live processing and monitoring resume in that block. |
| Silent passage during playback | Full processing continues; zero samples do not trigger ECO Stop. |

> **No extra wake delay or transport fade**
> Before the resumed block, old audio, filter and Ceiling histories are cleared and current settings applied. Existing Lookahead, oversampling latency and host PDC remain. The plugin adds no Stop/Start crossfade to its main output; leave monitoring fades to the DAW.

## FULL low-residual sleep and ASIO-Guard

FULL still processes live input and ordinary noise after Stop. It can suspend main audio work only on a known Stop without recording or offline rendering, when raw main and selected-sidechain peaks are at or below -180 dBFS, the actual gain-aware output has stayed below -160 dBFS for at least one second, and filter, delay and TP recovery are settled. Extremely low residue such as -200 dBFS becomes zero during sleep. Live input or selected-sidechain activity wakes processing in the current block. Extreme gain may conservatively keep FULL awake.

ASIO-Guard reflects the host's overall scheduling; per-plugin CPU time cannot be converted directly to its percentage. To check ECO at Stop, first confirm the host reports Stop. Export alignment still depends on host rendering and latency compensation.